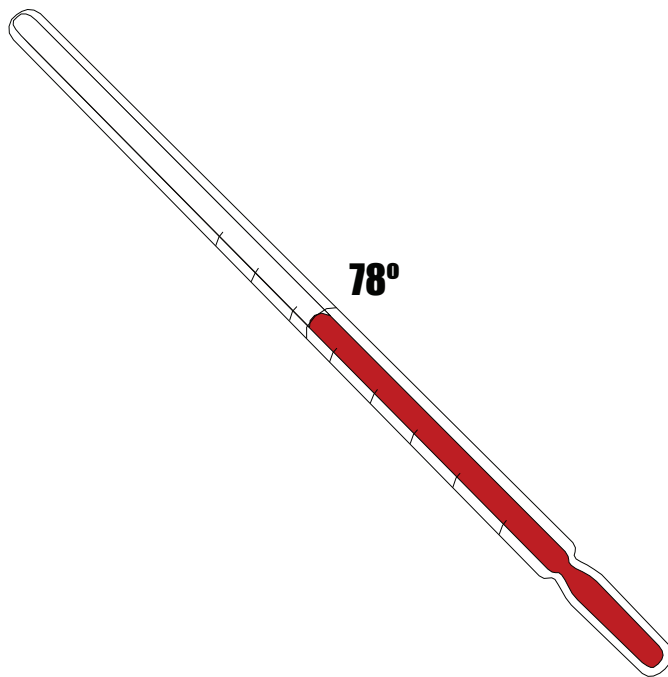


CHAPTER 7

REFRIGERATION



CHAPTER 7

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CHAPTER 7

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SECTION 7-1

REFRIGERATION SYSTEM

Your mixer may be equipped with a refrigeration jacket for control of final dough temperature. See **Chapter 2, Options Checklist**, to find out if your mixer has a refrigeration jacket.

This section is intended only as a general guideline for the design, installation and operation of a refrigeration system. We strongly recommend that you use only qualified refrigeration personnel for installation, maintenance and service of your refrigeration system.

⚠ DANGER

Liquid refrigerant can cause serious injury if it touches your skin. Always wear goggles, gloves and protective clothing over skin when working with any liquid refrigerant.

⚠ WARNING

Rupturing jacket or hoses can cause serious personal injury or death. Do not exceed a jacket pressure of 235 PSIG (16.2 bar).

⚠ WARNING

Refrigerant trapped in the jacket can cause very high pressures. Do not close shut off valves in both lines when liquid refrigerant is in jacket.

⚠ CAUTION

Correct installation and maintenance of your refrigeration system is required to insure trouble-free service. We strongly recommend that professional refrigeration personnel be used.

SECTION 7-1

SAFETY 7-1-1

▲ DANGER

Before servicing, repairing, or otherwise working on mixer, follow **LockOut/TagOut** procedure as required by **OSHA Standard 1910.147, The Control of Hazardous Energy**. See **Section 1-3, Zero Energy State Procedures**.

Follow these safety guidelines at all times when working on refrigeration systems.

- ✓ Protect your skin and eyes from contact with liquid refrigerant. Liquid refrigerant can cause skin irritation and can freeze skin on contact causing frostbite.
- ✓ Be sure all areas are well ventilated while working on refrigeration systems. Most refrigerants are nontoxic, but they will displace the air in a room causing suffocation.
- ✓ Avoid open flames and always work in a well-ventilated area. Fluorocarbon refrigerants will break down in a flame and produce dangerous gases. ***Never smoke while working on refrigeration systems.***
- ✓ Be sure all refrigerant is reclaimed from the refrigeration system and no pressure exists before opening the system.
- ✓ Use caution when handling refrigerant storage cylinders. Follow the cylinder provider's safety recommendations at all times.

SECTION 7-1

INDIRECT REFRIGERATION SYSTEMS 7-1-2

Peerless can supply your Double Arm mixer with an optional Indirect Refrigeration System.

An **Indirect Refrigeration System** uses cold water, glycol or brine. The cold liquid circulates through the mixer's bowl jacket. Heat transfers from the dough to the liquid and from the liquid to the refrigerant. This type system consists of a compressor, condenser, chiller, receiver, storage tank and pump. The evaporator is the heat exchanger for the liquid solution. The refrigerants used to cool the liquid are R-12, R-22, R502. However, because of the ozone-depleting potential of the chlorofluorocarbons (CFCs), Suva[®] 134a refrigerant is becoming more and more widely used as a replacement.

An ammonia system can be used in place of CFCs to cool the liquid flowing through the bowl jacket. Allow only experienced personnel to work on ammonia systems. Follow all installation and safety requirements provided by ammonia and compressor suppliers.

Under special conditions, hot water or steam can be used. Call the Engineering Department at **Peerless Machinery Corp.** to discuss the intended refrigeration cycle if you are using hot water or steam.

An **Indirect Refrigeration System** can be used with one mixer only or with a multiple mixer system.

SECTION 7-1

SIZING SYSTEMS 7-1-3

To properly size a refrigeration system, the heat load of the system must be calculated. As a service to our customers, **Peerless** will help make the necessary calculations as needed. To have the refrigeration requirements calculated for your system, fill out **Chart 7-1-1, Refrigeration Data Checklist**, and send to **Peerless Machinery Corp.**, Attn: Engineering Department. Your information will be entered into a custom program developed by **Peerless** and the output sent back. See **Figure 7-1-2, Refrigeration Data Sample Output**, for an example of a typical output.

Table 7-1-3 is intended as a guide to *approximate* a mixer refrigeration load. To calculate the *approximate* refrigeration load for your mixer, locate the correct drive motor high speed horsepower on the vertical axis of the chart. Draw a line straight across until it intersects the line that best describes your ingredient type. From this intersection, draw a vertical line down to the bottom axis. The value where the line intersects this axis equals the *approximate* refrigeration load for your mixer. For a more accurate refrigeration load calculation, fill out the **Refrigeration Data Checklist** discussed above.

INGREDIENT TEMPERATURES

To ensure a consistent dough-out temperature, the ingredient temperatures must be controlled where possible. In most cases, the flour temperature equals the ambient temperature, but the temperatures of water and other minor ingredients are controlled.

A heat exchanger usually controls water temperature. However, it is also common to mix the minor ingredients with the water in an ingrediator or tank, then cool the solution down to the desired temperature.

If you need assistance in designing, installing, or maintaining a refrigeration system for your mixer, we strongly recommend that you call Engineered Cold Systems (ECS) in Garden City, New York at (516) 437-7880. We consult with ECS for all our refrigeration needs; they have considerable experience with mixer refrigeration systems and will be very helpful.

SECTION 7-1

REFRIGERATION DATA CHECKLIST 7-1-1

CUSTOMER: _____

DATE: _____

COMPLETED BY: _____

Mixer Size and Type: _____

Drive Motor HP: _____

Product Type: _____

Low Mix Time, Min: _____

Coolant Type: _____

High Mix Time, Min: _____

Final Dough Temp: _____

REFRIG-ON Time, Min: _____

INGREDIENT DATA

	INGREDIENT	WEIGHT, LB.	STARTING TEMP, °F
1.	Flour	_____	_____
2.	Water	_____	_____
3.	Liquid Premix	_____	_____
4.	Sponge Dough	_____	_____
5.	Liquid Sponge	_____	_____
6.	Shortening	_____	_____
7.	Salt	_____	_____
8.	Oil	_____	_____
9.	Trace Solids	_____	_____
10.	Trace Liquids	_____	_____
11.	Ice	_____	_____
12.	Other _____	_____	_____
13.	Other _____	_____	_____

MISCELLANEOUS OTHER DATA

Summer Ambient Temp: _____

Winter Ambient Temp: _____

IF DX SYSTEM:

IF WATER SYSTEM:

Refrigeration Type: _____

Coolant Type: _____

Suction Pressure at Bowl: _____

Inlet Bowl Temp: _____

Suction Pressure at Comp: _____

Flow Rate, GPM: _____

SECTION 7-1

REFRIGERATION DATA SAMPLE OUTPUT CHART 7-1-2

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CUSTOMER: P E E R L E S S M A C H I N E R Y C O R P .

PRODUCT TYPE: BUN DOUGH

COOLANT TYPE: GLYCOL OR BRINE

LOW MIX TIME, MIN: 1.0

HIGH MIX TIME, MIN: 9.0

REFRIG ON TIME, MIN: 9.0

REFRIGERATION CALCULATIONS AND DATA

SIZE AND TYPE OF MIXER: HS24

TYPE OF DOUGH: SPONGE & DOUGH

DRIVE MOTOR HP: 125.0

AVG MOTOR PULL, %: 90.0

JACKET AREA, FT²: 62.3

SUCTION PRESSURE, PSI: N/A

USER: marc

INGREDIENT	SPECIFIC HEAT CONST. BTU/LB F	WEIGHT LBS	STARTING TEMP DEGREE F	HEAT CONTENT BTU	% OF TOTAL
1 FLOUR	0.42	330.0	90.0	1,663.2	4.12
2 WATER	1.00	220.0	34.0	-9,680.0	-23.97
3 LIQUID PREMIX	0.90	0.0	0.0	0.0	0.00
4 SPONGE DOUGH	0.60	1300.0	85.0	5,460.0	13.52
5 LIQUID SPONGE	0.70	0.0	0.0	0.0	0.00
6 SHORTENING	0.50	0.0	0.0	0.0	0.00
7 SALT	0.30	22.0	80.0	13.2	0.03
8 OIL	0.60	8.0	90.0	57.6	0.14
9 TRACE SOLIDS	0.50	28.0	80.0	28.0	0.07
10 TRACE LIQUIDS	0.90	0.0	0.0	0.0	0.00
11 ICE	1.00	0.0	0.0	0.0	0.00
12 OTHER (YEAST)	0.60	8.0	34.0	-211.2	-0.52
13 OTHER (SUGAR)	0.30	172.0	80.0	103.2	0.26
TOTAL, INGREDIENTS		2088.0		-2,566.0	-6.35
14 HEAT OF HYDRATION OF FLOUR		330.0		2,145.0	5.31
15 LATENT HEAT OF FUSION OF ICE		0.0		0.0	0.00
16 HEAT GENERATED BY MOTOR				40,812.4	101.04
TOTAL HEAT CONTENT, BTU				40,391.4	

DESCRIPTION	*STANDARD RESULTS	
	OPTION 1	OPTION 2
17 FINAL DOUGH TEMP, F	78.0	74.4
18 HEAT TRANSFER RATE, BTU/HR	269,276	269,276
19 HEAT TRANSFER RATE, TONS	22.4	22.4
20 OVERALL THERMAL HEAT COEF, BTU/HR FT ² F	66.2	70.0
21 AVG JACKET TEMP, F	12.7	12.7
22 INLET TEMP, F	10.0	10.0
23 FLOW RATE, GPM	100.0	100.0
24 OUTLET TEMP, F	15.4	15.4

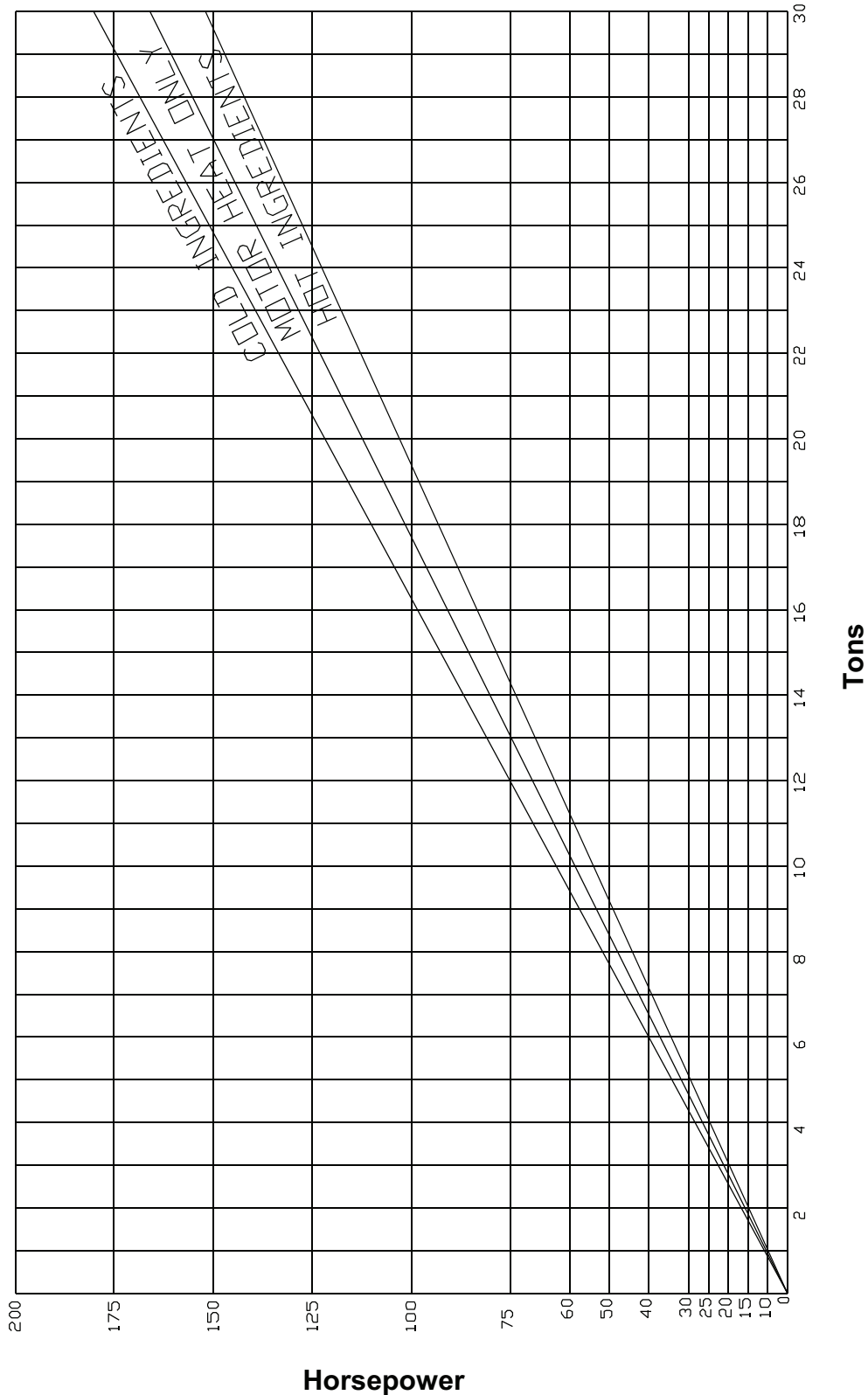
* OPTION 1 ASSUMES THE FINAL DOUGH TEMP AND HEAT TRANSFER RATE ARE THE SAME AS INPUTED AND CALCULATES WHAT THE JACKET TEMP SHOULD BE USING THE EXPECTED OVERALL THERMAL HEAT COEFFICIENT.

* OPTION 2 ASSUMES THE JACKET TEMP AND HEAT TRANSFER RATE ARE THE SAME AS INPUTED AND CALCULATES WHAT THE DOUGH TEMP SHOULD BE USING THE EXPECTED OVERALL THERMAL HEAT COEFFICIENT.

NOTE! ALL THE ABOVE DATA AND RESULTS ARE THEORETICAL AND ARE NOT TO BE CONSTRUED AS GUARANTEES. YOUR ACTUAL RESULTS MIGHT VARY FROM FIGURES SHOWN HERE.

SECTION 7-1

MIXER MOTOR--TONNAGE VS HORSEPOWER CHART 7-1-3



NOTE: For BTU/HR, multiply X 12,000.

SECTION 7-1

Table 7-1-4 shows typical refrigeration system specifications. **Peerless** intends the information on this table only as a general guideline for the specifications of a mixer refrigeration system. We strongly recommend that you allow only experienced, qualified refrigeration personnel to calculate the specific refrigeration requirements of your mixer.

TYPICAL REFRIGERATION SYSTEM SPECIFICATIONS TABLE 7-1-4

INDIRECT REFRIGERATION SYSTEM	AREA	SPECIFICATION
COLD WATER	Inlet water temperature	35° F (2° C)
	Outlet water temperature	40° F (4° C)
	Flow rate	See Table 7-1-6
GLYCOL	Inlet glycol temperature	25° F (-4° C)
	Outlet glycol temperature	30° F (-1° C)
	Flow rate	See Table 7-1-6
	Glycol solution	30%

GLYCOL SOLUTION RECOMMENDATIONS TABLE 7-1-5

Peerless recommends DOWFROST[®], an inhibited propylene glycol-based fluid produced by the Dow Chemical Company and chemically approved by the U.S. Department of Agriculture (USDA) for creating a glycol solution.

Keep in mind that the strength of the glycol should be about 5° lower than the evaporating temperature of the refrigerant. The evaporating temperature of the refrigerant should be 10° lower than the inlet temperature.

The following table lists the recommended glycol percentage for different temperatures.

SECTION 7-1

NOTE: A glycol solution with a stronger or weaker percentage of glycol than recommended for the temperature used will adversely affect the glycol's heat transfer capability.

INLET TEMPERATURE DESIRED F (C)	FREEZE POINT OF GLYCOL SOL. F (C)	% GLYCOL	% WATER
0° (-18°)	-15° (-26°)	45	55
5° (-15°)	-10° (-23°)	42	58
10° (-12°)	-5° (-21°)	40	60
15° (-9°)	0° (-18°)	37	63
20° (-7°)	5° (-15°)	35	65
25° (-4°)	10° (-12°)	30	70
30° (-1°)	15° (-9°)	25	75

INDIRECT REFRIGERATION RECOMMENDED FLOW RATES TABLE 7-1-6

MIXER SIZE	WITHOUT BEC	
	FLOW RATE Gallons per minute (Liters per min.)	*PRESSURE DROP PSI (Bar)
DA100/150	30-50 (114-189)	10 (0.7)
DA200/250	40-60 (151-227)	15 (1.0)
DA300/350	50-70 (189-265)	20 (1.4)
DA400/500	60-80 (227-303)	25 (1.7)
DA600/650	70-90 (265-341)	30 (2.1)

NOTE: * Pressure Drop figures are based on 35% glycol at 25° F (-4° C) and are approximately ± 10 PSI.

SECTION 7-2

INSTALLATION OF REFRIGERATION SYSTEMS

NOTE: Use only qualified personnel for design and installation of your refrigeration system.

Peerless strongly suggests you use a contractor who is experienced in **mixer refrigeration systems**. Keep in mind that even though contractors may be very familiar with standard air conditioning or blast freezers, they may *not* have experience with mixer refrigeration systems. Mixer refrigeration systems are unique in that they have widely varying heat loads. Your system must be designed and installed to allow for these varying heat loads.

Because every **refrigeration system** has unique requirements, each system must be designed and installed according to each specific mixer application. However, **Peerless** suggests the following as general guidelines:

SECTION 7-2

INDIRECT REFRIGERATION SYSTEMS 7-2-1

See **Figure 7-2-1, Indirect Refrigeration System Schematic**. The schematic shows two mixers using the refrigeration package. It is very common and even desirable to have multiple heat loads, for example, operating several mixers and water chillers using the same glycol refrigeration package. This type system must be designed per application, therefore, will not be discussed in this manual.

Peerless supplies and ships loose with each mixer the refrigeration hoses with fittings. See **Figure 3-6-1, Refrigeration Hose Locations**, for the hose and connection sizes for all mixer models. See **Section 3-7** for instructions for installing hoses.

Typically, an **indirect refrigeration system** is controlled by starting or stopping a pump motor. An alternate method is to open and close a shut-off valve (not supplied by **Peerless**). If you use the shut-off valve method, be sure the valve or pump has a bypass feature for the liquid coolant.

For shut-off valve refrigeration control method, wire the shut-off valve solenoid per the mixer's wiring diagram. Connect the mixer's neutral wire to one side of the coil, and connect the other side to the control wire for the liquid line solenoid as shown on the wiring diagram. The wires for the shut-off valve solenoid are located either on the terminal strip in the starter enclosure or on the terminal strip in the rear junction box of the mixer. The maximum current draw for the valve is 2 amps. Also, be sure the voltage of the valve matches the control voltage of the mixer.

If a pump motor is used, wire the starter coil for the pump motor the same way as the shut-off valve solenoid above.

Install hand shut-off valves in the inlet and outlet lines close to the hose connections. Hand shut-off valves are not supplied by **Peerless**.

SECTION 7-2 **CAUTION**

Never close valves in inlet and outlet lines at the same time unless hoses are being changed. Trapped glycol in the bowl will cause high pressure in jacket resulting in damage.

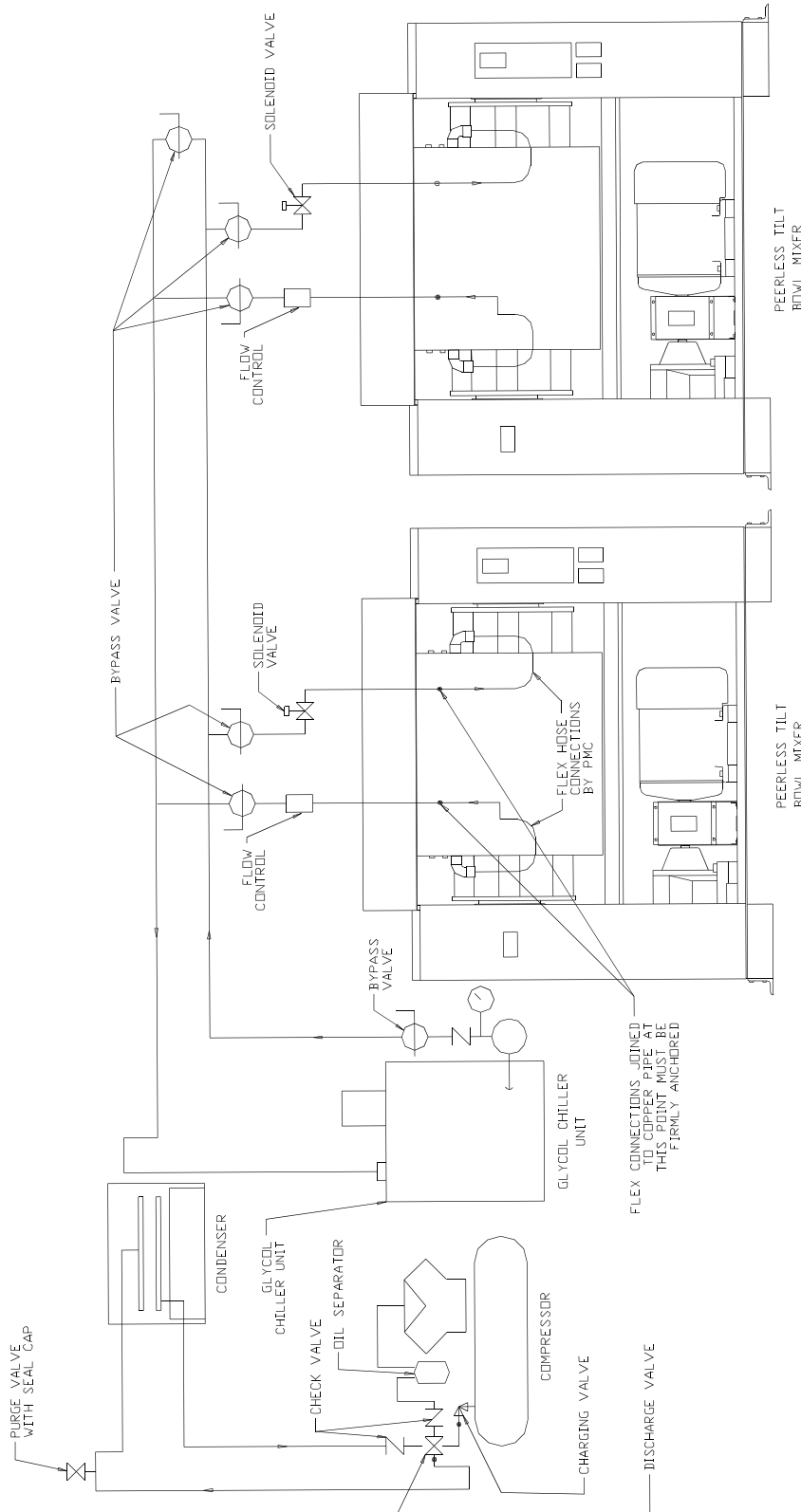
Peerless recommends that you install a flow meter in the inlet line to the mixer to make trouble-shooting the **refrigeration system** easier and more reliable. The flow meter is not supplied by **Peerless**.

We also recommend that you install thermometers in the inlet and outlet lines next to the mixer to help with trouble-shooting the **refrigeration system**. These thermometers are not supplied by **Peerless**.

If you need help in designing and installing your **refrigeration system**, call the Engineering Department at **Peerless Machinery Corp.** In addition, we strongly recommend that you call Engineered Cold Systems (ECS) in Garden City, New York at (516) 437-7880. We use ECS as a consultant for all our refrigeration needs; they have considerable experience with mixer refrigeration systems.

SECTION 7-2

INDIRECT REFRIGERATION SYSTEM SCHEMATIC FIGURE 7-2-1



NOTE: Use schematic as reference only. Not to be used for construction.

SECTION 7-3

GENERAL SPECIFICATIONS

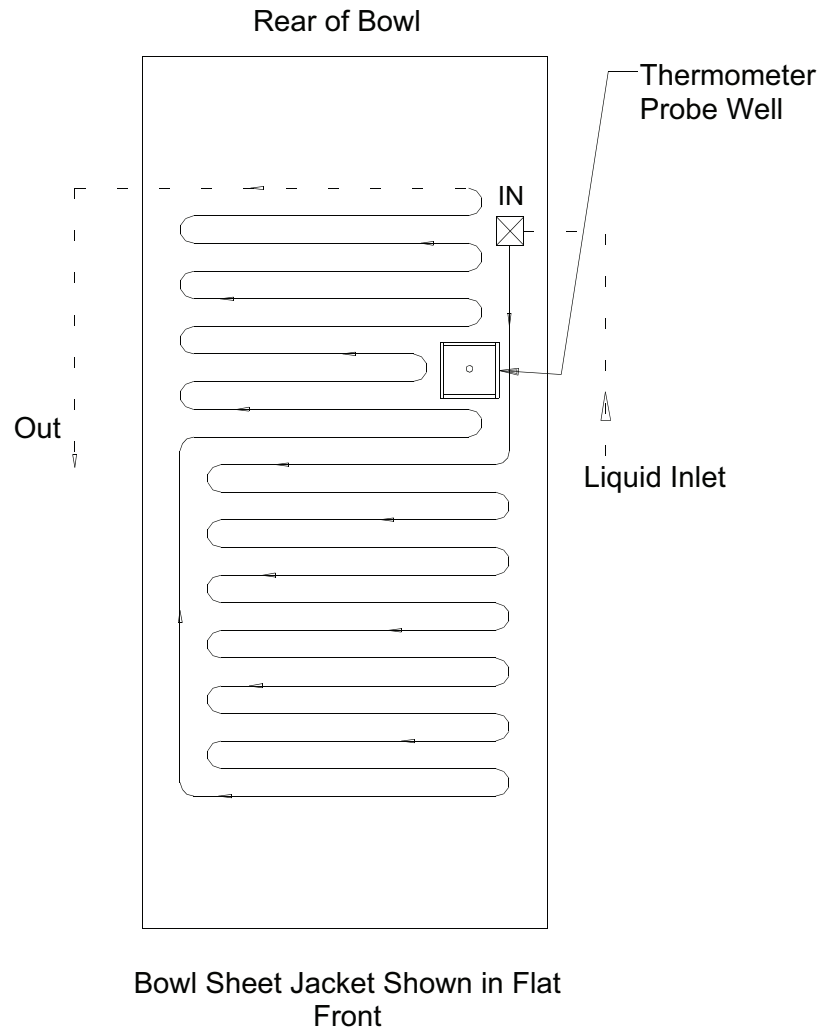
The following information is intended as a general guideline covering general specifications for your mixer's refrigeration system. If you need more specific information or calculations, call the Engineering Department at **Peerless Machinery Corp.**

JACKET VOLUME TABLE 7-3-1

MIXER SIZE	BOWL SHEET JACKET CAPACITY Gallons (Liters)
DA100	5.2 (19.5)
DA150	6.9 (26.2)
DA200/250	10.4 (39.4)
DA300/350	14.4 (54.4)
DA400	17.0 (64.3)
DA500	18.9 (71.4)
DA600/650	22.2 (84.1)

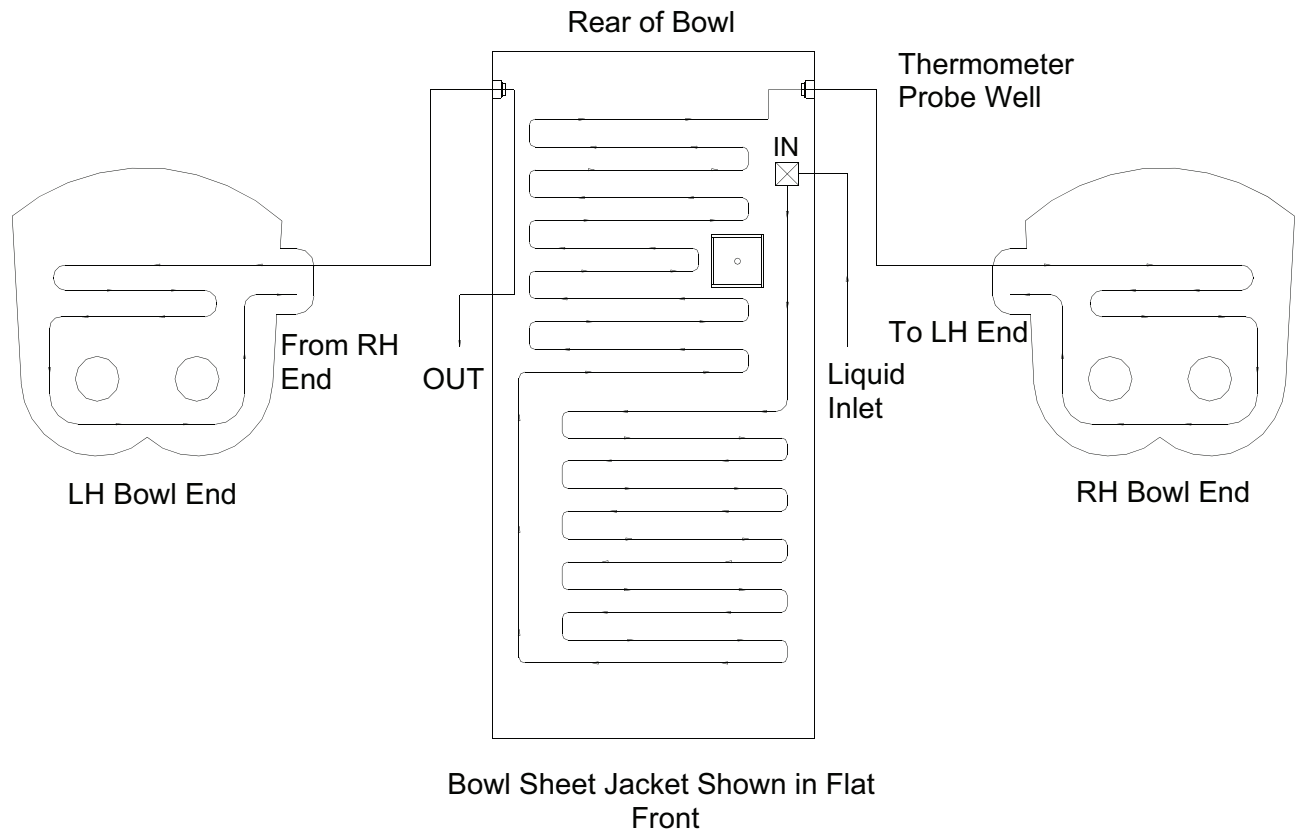
SECTION 7-3

STANDARD BOWL COOLANT FLOW FIGURE 7-3-2



SECTION 7-3

BOWL COOLANT FLOW WITH END COOLING FIGURE 7-3-3



SECTION 7-4

REFRIGERATION SYSTEM TROUBLE-SHOOTING GUIDE

Table 7-4-1 is designed to be a quick reference for trouble spots that may arise with your mixer's refrigeration system. This guide explains at a glance the trouble, possible causes and remedies.

For further assistance, call the Service Department at **Peerless Machinery Corp.** We also recommend that you call Engineered Cold Systems (ECS) in Garden City, New York at (516) 437-7880. We consult with ECS for all our refrigeration needs; they have considerable experience with mixer refrigeration systems and will be very helpful.

NOTE: The Environmental Protection Agency (EPA) requires refrigerant recovery. <i>Use only certified technicians, qualified to do recovery work.</i>
--

SECTION 7-4

OPTIONAL REFRIGERATION CONTROLS TROUBLE-SHOOTING GUIDE

TABLE 7-4-1

TROUBLE	CAUSE	REMEDY
Refrigeration will not come on (liquid cooling jacket). REFRIGERATION ON light flashing.	Suction line thermostat (DX jackets only) not jumpered out.	Add jumper. See wiring diagram.
Refrigeration will not come on. REFRIGERATION ON light not flashing.	REFRIGERATION ON timer not set.	Set time on refrigeration timer.
	REFRIGERATION SELECTOR switch in OFF position.	Select either ON or AUTO position.
	Refrigeration system turned off.	Turn on.
	Refrigeration system faulty.	Locate fault and repair.
Dough temperature is too cold.	Refrigeration left on too long.	Decrease time refrigeration is on.
	REFRIGERATION SELECTOR is in ON position.	Move to AUTO position.
Dough temperature is too hot.	Refrigeration not on long enough.	Increase time refrigeration is on.
	Refrigeration system not working.	See above.
	Ice forming inside bowl.	Delay start of refrigeration cycle.
		Increase temperature of refrigerant.
Refrigeration is on. No frost pattern.	Faulty bowl jacket.	Call Peerless Service Dept.
	Faulty refrigeration system.	Locate problem and repair.